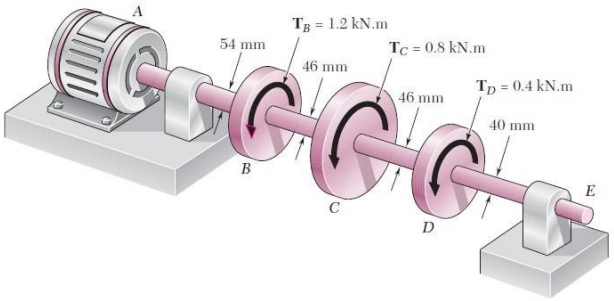
Question 1.
Under normal operating conditions, the electric motor exerts a torque of 2.4 kNm at A. Knowing that each shaft is solid, determine the maximum shearing stress (a) in shaft AB, (b) in shaft BC, (c) in shaft CD.



SOLUTION

Shaft AB: $T_{AB} = 2.4 \times 10^3 \text{ N} \cdot \text{m}, \qquad c = \frac{1}{2}d = 0.027 \text{ m}$

$$\tau_{AB} = \frac{Tc}{J} = \frac{2T}{\pi c^3} = \frac{(2)(2.4 \times 10^3)}{\pi(0.027)^3} = 77.625 \times 10^6 \text{ Pa} = 77.6 \text{ MPa}$$ ◄

Shaft BC: $T_{BC} = 2.4 \text{ kN} \cdot \text{m} - 1.2 \text{ kN} \cdot \text{m} = 1.2 \text{ kN} \cdot \text{m}, \qquad c = \frac{1}{2}d = 0.023 \text{ m}$

$$\tau_{BC} = \frac{Tc}{J} = \frac{2T}{\pi c^3} = \frac{(2)(1.2 \times 10^3)}{\pi(0.023)^3} = 62.788 \times 10^6 \text{ Pa} = 62.8 \text{ MPa}$$ ◄

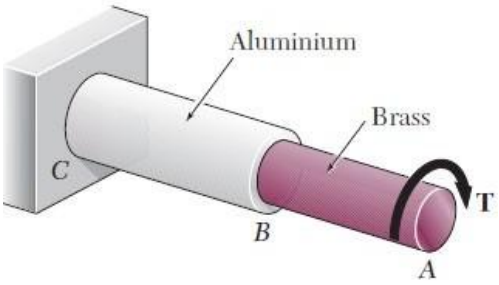
Shaft CD: $T_{CD} = 0.4 \times 10^3 \text{ N} \cdot \text{m} \qquad c = \frac{1}{2}d = 0.023 \text{ m}$

$$\tau_{CD} = \frac{Tc}{J} = \frac{2T}{\pi c^3} = \frac{(2)(0.4 \times 10^3)}{\pi(0.023)^3} = 20.929 \times 10^6 \text{ Pa} = 20.9 \text{ MPa}$$ ◄

Question 2.

Will be answered in the Lecture-tute.

The allowable stress is 50 MPa in the brass rod AB and 25 MPa in the aluminium rod BC. Knowing that a torque T = 1250 Nm is applied at A, determine the required diameter of (a) rod AB, (b) for BC.



SOLUTION

$$\tau_{\max} = \frac{Tc}{J} \qquad J = \frac{\pi}{2}c^4 \qquad c^3 = \frac{2T}{\pi\tau_{\max}}$$

Rod AB:
$$c^3 = \frac{(2)(1250)}{\pi(50 \times 10^6)} = 15.915 \times 10^{-6}\ \text{m}^3$$

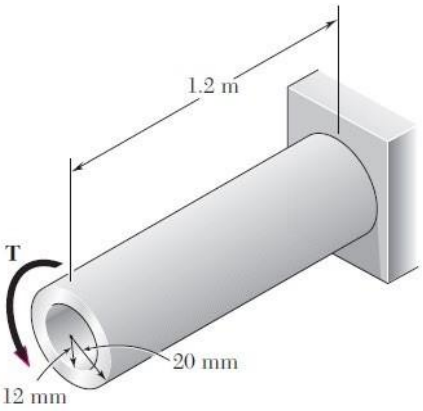
$$c = 25.15 \times 10^{-3}\ \text{m} = 25.15\ \text{mm} \qquad d_{AB} = 2c = 50.3\ \text{mm}$$

Rod BC:
$$c^3 = \frac{(2)(1250)}{\pi(25 \times 10^6)} = 31.831 \times 10^{-6}\ \text{m}^3$$

$$c = 31.69 \times 10^{-3}\ \text{m} = 31.69\ \text{mm} \qquad d_{BC} = 2c = 63.4\ \text{mm}$$

Question 3.

For the aluminium shaft show (G = 27 GPa), determine (a) the torque T which causes an angle of twist of 5°, (b) the angle of twist caused by the same torque T in a solid cylindrical shaft of the same length and cross-sectional area.



SOLUTION

(a) $\varphi = \dfrac{TL}{GJ}$, $\qquad T = \dfrac{GJ\varphi}{L}$

$\varphi = 5° = 87.266 \times 10^{-3}$ rad,

$L = 1.2$ m

$J = \dfrac{\pi}{2}(c_2^4 - c_1^4)$

$= \dfrac{\pi}{2}(0.020^4 - 0.012^4) = 0.218 \times 10^{-6}\ \text{m}^4$

$T = \dfrac{(27 \times 10^9)(0.218 \times 10^{-6})(87.266 \times 10^{-3})}{1.2}$

$= 428\ \text{N} \cdot \text{m}$

(b) Hollow shaft $\qquad A = \pi(c_2^2 - c_1^2)$ $\qquad$ Solid shaft $= A = \pi c^2$

Matching areas $\qquad c^2 = c_2^2 - c_1^2 = 0.020^2 - 0.012^2 = 0.000256\ \text{m}^2$

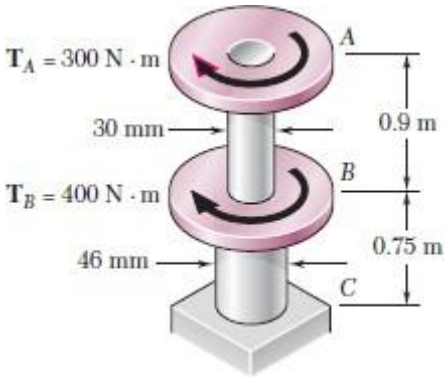
$c = 0.016$ m, $\qquad J = \dfrac{\pi}{2}c^4 = \dfrac{\pi}{2}(0.016)^4 = 0.102 \times 10^{-6}\ \text{m}^4$

$\varphi = \dfrac{TL}{GJ} = \dfrac{(428)(1.2)}{(27 \times 10^9)(0.102 \times 10^{-6})} = 186.49 \times 10^{-3}$ rad

$= 10.68°$

Question 4.

The torques shown are exerted on pulleys A and B. Knowing that the shafts are solid and made of aluminium (G = 77 GPa), determine the angle of twist between (a) A and B, (b) A and C.



SOLUTION

(a) $T_{AB} = 300\ \text{N} \cdot \text{m}, \qquad L_{AB} = 0.9\ \text{m}, \qquad c_{AB} = \frac{1}{2} d =$ 0.015m

$$J_{AB} = \frac{\pi}{2} (0.015)^4 = 79.522 \times 10^{-9}\ \text{m}^4$$

$$\varphi_{AB} = \frac{T_{AB} L_{AB}}{GJ} = \frac{(300)(0.9)}{(77 \times 10^9)(79.522 \times 10^{-9})} = 44.095 \times 10^{-3}\ \text{rad}$$

$$\varphi_{AB} = 2.53^\circ \quad \blacktriangleleft$$

(b) $T_{BC} = 300 + 400 = 700\ \text{N} \cdot \text{m}, \qquad L_{BC} = 0.75\ \text{m}, \qquad C_{BC} = \frac{1}{2} d = 0.023\ \text{m}$

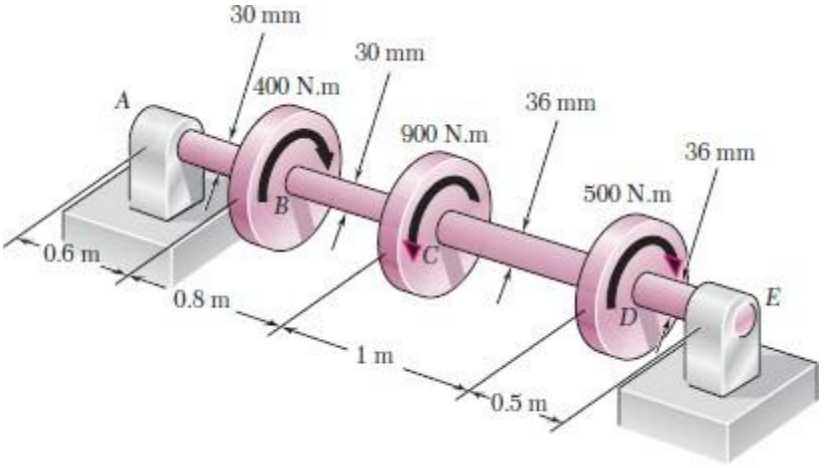
$$J_{BC} = \frac{\pi}{2} (0.023)^4 = 439.573 \times 10^{-9}\ \text{m}^4$$

$$\varphi_{BC} = \frac{T_{BC} L_{BC}}{GJ_{BC}} = \frac{(700)(0.75)}{(77 \times 10^9)(439.573 \times 10^{-9})} = 15.511 \times 10^{-3}\ \text{rad}$$

$$\varphi_{AC} = \varphi_{AB} + \varphi_{BC} = 59.606 \times 10^{-3}\ \text{rad} = 3.42^\circ \quad \blacktriangleleft$$

Question 5.
Will be answered in the Lecture-tute.

The torque shown are exerted on pulleys B, C and D. Knowing that the entire shaft is made of steel (G = 27 GPa), determine the angle of twist between (a) C and B, (b)D and B.



SOLUTION

(a) Shaft BC:
$$c = \frac{1}{2}d = 0.015 \text{ m}$$
$$J_{BC} = \frac{\pi}{4}c^4 = 79.522 \times 10^{-9} \text{ m}^4$$
$$L_{BC} = 0.8 \text{ m}, \qquad G = 27 \times 10^9 \text{ Pa}$$
$$\varphi_{BC} = \frac{TL}{GJ} = \frac{(400)(0.8)}{(27 \times 10^9)(79.522 \times 10^{-9})}$$
$$= 0.149904 \text{ rad} = 8.54^\circ$$

(b) Shaft CD:
$$c = \frac{1}{2}d = 0.018 \text{ m} \qquad J_{CD} = \frac{\pi}{4}c^4 = 164.896 \times 10^{-9} \text{ m}^4$$
$$L_{CD} = 1.0 \text{ m} \qquad T_{CD} = 400 - 900 = -500 \text{ N} \cdot \text{m}$$
$$\varphi_{CD} = \frac{TL}{GJ} = \frac{(-500)(1.0)}{(27 \times 10^9)(164.896 \times 10^{-9})} = -0.11230 \text{ rad}$$
$$\varphi_{BD} = \varphi_{BC} + \varphi_{CD} = 0.14904 - 0.11230 = 0.03674 \text{ rad} = 2.11^\circ$$

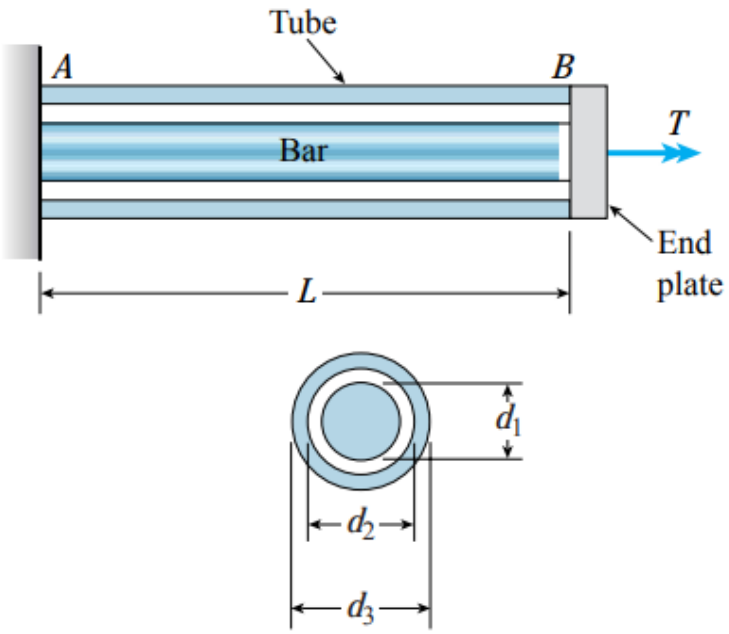
Question 6.

A solid steel bar of diameter d_1 = 25.0 mm is enclosed by an aluminum tube of outer diameter d_3 = 37.5 mm and inner diameter d_2 = 30.0 mm. Both bar and tube are held rigidly by a support at end A and joined securely to a rigid plate at end B. The composite bar, which has a length L = 550 mm, is twisted by a torque T = 400 Nm acting on the end plate.

The shear modulus of the steel is G_{st} = 80 GPa, the shear modulus of the aluminum is G_{al} = 28 GPa.

(a) Determine the angle of rotation ϕ (in degrees) of the end plate.

(b) Determine the maximum shear stresses τ_1 and τ_2 in the bar and tube, respectively.



Solutions

$$J_1 = \frac{\pi d_1^4}{32} = 38349.52 \text{ mm}^4, \quad J_2 = \frac{\pi(d_3^4 - d_2^4)}{32} = 114622.88 \text{ mm}^4$$

$$\phi = \frac{TL}{G_{st}J_1 + G_{al}J_2} = \frac{400{,}000 \times 550}{80{,}000 \times 38349.52 + 28{,}000 \times 114622.88} = 0.03506 \text{ rad} = 2.008^\circ$$

$$\tau_1 = \frac{T_1(d_1/2)}{J_1} = \frac{100.23(1000)(25/2)}{38349.52} = 63.72 \text{ MPa}$$

$$\tau_2 = \frac{T_2(d_3/2)}{J_2} = \frac{299.72(1000)(37.5/2)}{114622.88} = 33.45 \text{ MPa}$$